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Mr Kiser is our primary stakeholder because he is the one having the problem at hand. During the class interview, Mr Kiser has said that these are some of the top requirements he wants in our handles. Starting with the main objective, Mr. Kiser requested that the final handle extend the height of the original handle to reduce the amount that he has to bend over. After the interview, our team created a list of design requirements including the ability to be easily attachable, cheaper, ergonomically fit, lightweight, aesthetically pleasing, and adjustable.

To dive deeper into the requirements and criteria…

1. **Adjustable**: An adjustable handle should offer a range of settings to accommodate people of all different heights. Making sure the adjustment process is smooth and user-friendly is also key.
2. **Durability:** Having a product that is durable and can last long term will save money and time. The strength and longevity is a reflection of the time and effort spent by us, the team members. Durability reduces waste and in turn protects the environment.
3. **Cheap**: Cost-effectiveness can be reached by using affordable, durable materials like plastic or aluminum. However, in this case it was achieved by using free materials given by the robotics club. Additionally, a design that uses fewer parts or standardized components can help keep the overall costs down. Considering the overall costs, a more simple design would be able to achieve this goal more efficiently.
4. **Lightweight**: Using lightweight materials such as aluminum or plastic materials can keep the handle light without causing the strength or durability of the handle to fluctuate. A lightweight handle enhances the overall mobility of the cart, making it easier to push and turn, especially when the cart is full, in this case with ACT books.

5. **Ergonomically Fit**: The handle's design should follow ergonomic principles to ensure comfort when using. This includes features like a handle that fits the natural shape of the hand, soft-touch materials, and a width that distributes pressure evenly. Also width that can accommodate people with larger or smaller shoulder widths. Ergonomic design can reduce strain and fatigue on the body, making the cart easier to handle, especially over long periods of time.
6. **Easily Attachable**: The handle should feature an easy mechanism, such as a basic clip or a snap-in system, ideally requiring minimal effort and tools to attach. Another important aspect of attachment is that it should not require more than one person to put onto the cart. This mechanism should be simple, so the consumer can easily understand how to attach and detach the handle without having to refer to an instruction manual.
7. **Aesthetically Pleasing**: The handle should have a visually appealing design, possibly using sleek lines and a more simple, modern look. Color choices and textures can also affect its attractiveness. A well-designed handle can add to the overall appeal of the cart, encouraging users to choose it over less aesthetic options.

These components discuss the essential aspects of each requirement. Making a cart handle that is user-friendly, cost-effective, and aesthetically pleasing is key to having a good product.